

Spring Framework Exercises Report

Exercise 1 & 2 - Basic Spring + Dependency Injection

Source Code:

```
pom.xml:
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.30</version>
</dependency>

applicationContext.xml:
<bean id="bookRepository" class="com.library.repository.BookRepository"/>
<bean id="bookService" class="com.library.service.BookService">
  <property name="bookRepository" ref="bookRepository"/>
</bean>

BookRepository.java
public class BookRepository{
  public void display(){System.out.println("Book Repository");}
}

BookService.java
public class BookService{
  private BookRepository bookRepository;
  public void setBookRepository(BookRepository b){bookRepository=b;}
  public void service(){System.out.println("Book Service");bookRepository.display();}
}
```

Expected Output:

Book Service
Book Repository

Exercise 4 - Maven Project

Source Code:

```
pom.xml:
spring-context
spring-aop
spring-webmvc

<plugin>
<maven-compiler-plugin>
<source>1.8</source>
<target>1.8</target>
</plugin>
```

Expected Output:

BUILD SUCCESS

Exercise 5 & 7 - IoC, Constructor and Setter Injection

Source Code:

```
Constructor:
public BookService(BookRepository repo){this.bookRepository=repo;}

XML:
<constructor-arg ref="bookRepository"/>
<property name="bookRepository" ref="bookRepository"/>
```

Expected Output:

Constructor Injection Successful Setter Injection Successful

Exercise 9 - Spring Boot CRUD

Source Code:

```
application.properties
spring.datasource.url=jdbc:h2:mem:testdb
```

```
@Entity
class Book{ @Id @GeneratedValue Long id; String title; }

interface BookRepository extends JpaRepository<Book,Long>{}

@RestController
@RequestMapping("/books")
class BookController{
    @Autowired BookRepository repo;
    @GetMapping List<Book> all(){return repo.findAll();}
    @PostMapping Book add(@RequestBody Book b){return repo.save(b);}
}
```

Expected Output:

GET /books -> []

POST /books -> Book Saved

GET /books -> [{id:1,title:'Java'}]